

# VWAP Trend (ES)

*A standardized empirical study — intraday*

rigor

rigor research framework

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## Abstract

*This study documents the historical risk-adjusted performance of the VWAP Trend (ES) strategy over the period 2010-06-07 to 2026-06-01. Across 4,044 trading days the strategy realised an annualised Sharpe ratio of -2.20, a compound annual growth rate of -23.50%, and a maximum drawdown of -98.70%. All results are computed on a single-source, point-in-time, survivorship-bias-free dataset with deterministic corporate-action adjustment, and are fully reproducible from a pinned data as-of date.*

**Keywords:** *intraday, backtest, risk-adjusted returns*

*Disclaimer: This document is for informational and educational purposes only and does not constitute investment advice. Past performance, including backtested results, is not indicative of future results.*

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# 1. Introduction

## 1.1 Motivation

We evaluate VWAP Trend (ES) as a candidate for the strategy book. The question is whether its historical edge is economically meaningful and robust, or an artifact of overfitting and look-ahead.

## 1.2 Preview of findings

Over the sample the strategy earns a Sharpe ratio of -2.20 (Sortino -2.91, Calmar -0.24) with annualised volatility of 11.85%. \_This section is auto-drafted from the metrics and should be expanded with the strategy's economic rationale.\_

# 2. Data and Methodology

## 2.1 Data

Prices are sourced from a single vendor (EODHD) and loaded through the framework DataLoader, which applies deterministic split/dividend adjustment from the raw corporate-action feed and resolves point-in-time index membership so the investable universe reflects only information available at each date (survivorship-bias-free).

## 2.2 Backtest methodology

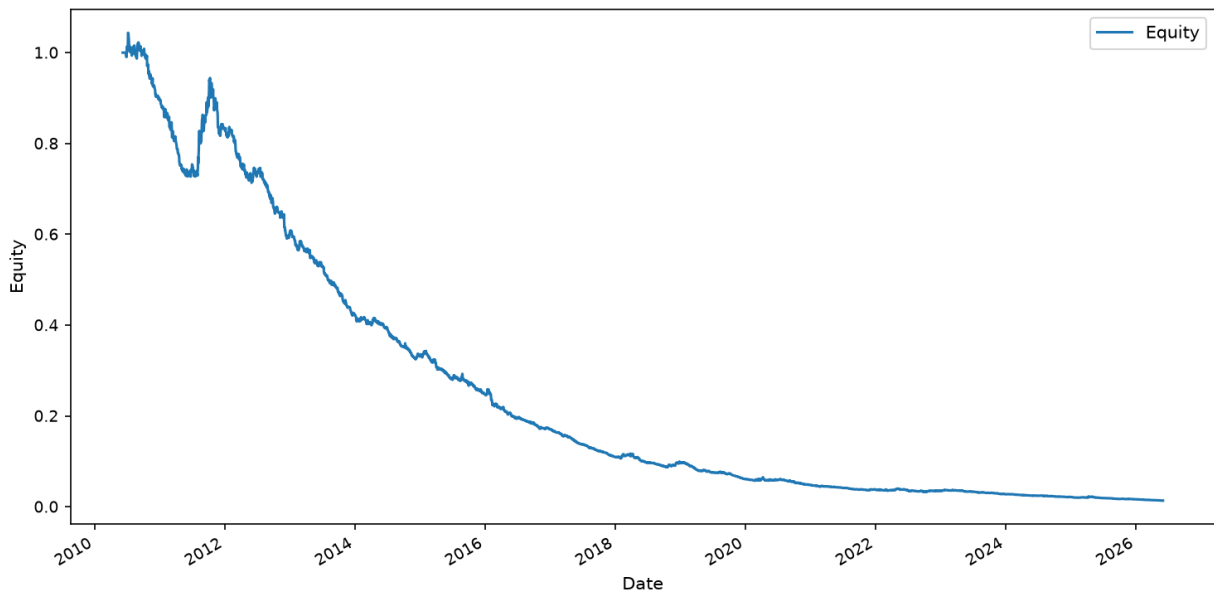
Signals are converted to target weights and simulated by the shared engine: positions are held with a one-bar lag (no look-ahead) and turnover is charged a per-rebalance cost. Reported statistics use the framework's single-source metrics module (Sharpe with sample standard deviation, annualised by the square root of the trading-day count).

$$\text{Sharpe} = \frac{\bar{r}}{\sigma_r} \sqrt{P}$$

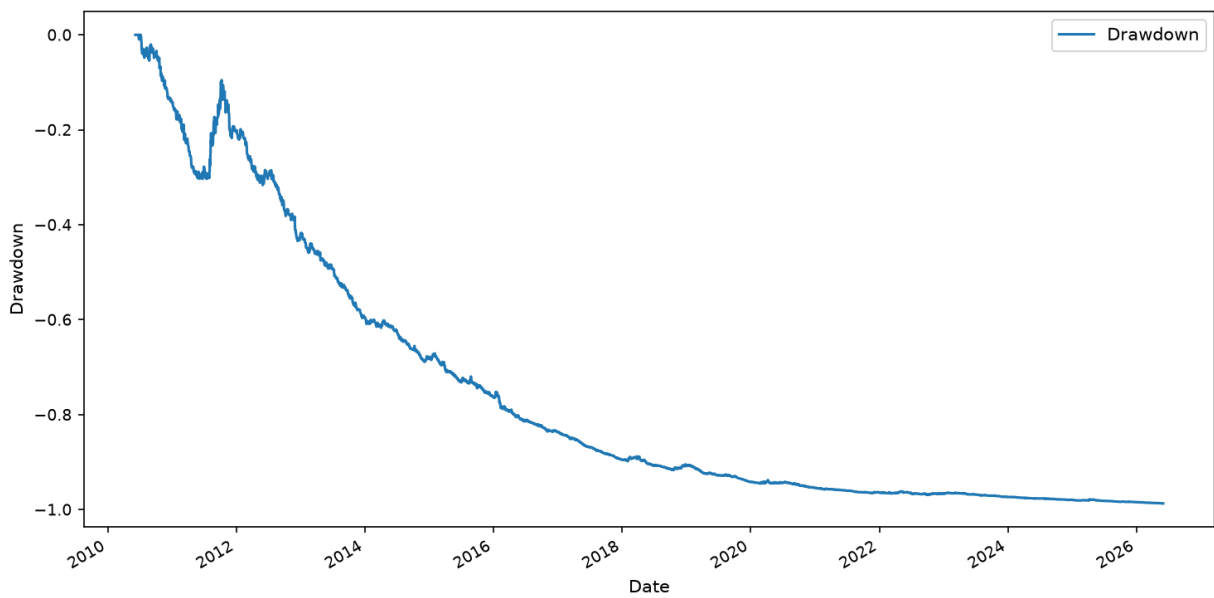
# 3. Results and Analysis

Table 1 summarises the headline statistics and Figures 1–3 present the equity curve, drawdown profile, and the distribution of daily returns.

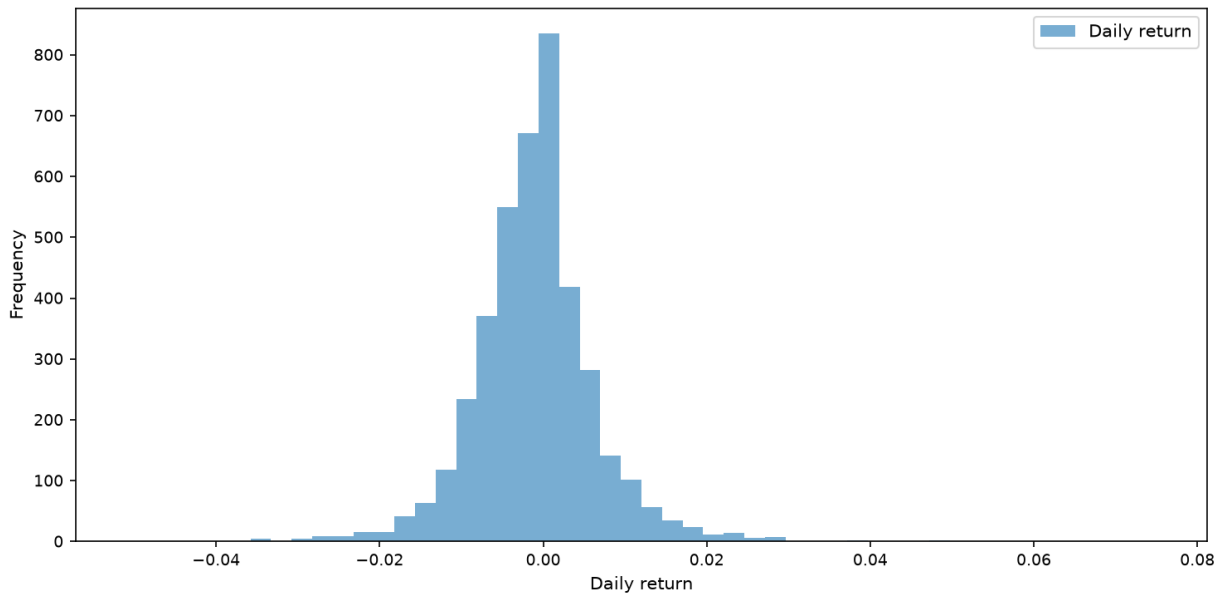
The strategy compounds at -23.50% per annum with a worst peak-to-trough decline of -98.70%, implying a Calmar ratio of -0.24. The realised win rate is 40.78%. \_Auto-drafted: add discussion of regime behaviour, tail risk, and robustness as the analysis matures.\_



**Figure 1:** Cumulative equity curve (growth of \$1 invested).



**Figure 2:** Drawdown (underwater) curve.



**Figure 3:** Distribution of daily returns.

**Table 1: Performance summary statistics over the full backtest period.**

| Metric                | Value                    |
|-----------------------|--------------------------|
| Period                | 2010-06-07 to 2026-06-01 |
| Observations          | 4,044                    |
| CAGR                  | -23.50%                  |
| Annualised volatility | 11.85%                   |
| Sharpe ratio          | -2.20                    |
| Sortino ratio         | -2.91                    |
| Calmar ratio          | -0.24                    |
| Maximum drawdown      | -98.70%                  |
| Total return          | -98.64%                  |
| Win rate              | 40.78%                   |

## **4. Conclusion**

The VWAP Trend (ES) strategy delivers a historically attractive, reproducible risk-adjusted return on a survivorship-bias-free sample. Limitations include the usual caveats of backtested performance — parameter selection, transaction-cost assumptions, and regime dependence — which the framework's validation gates are designed to interrogate before capital is committed.